

UCH-ADZHI, Turkmenistan

WMO: 387990

Lat: 38.083N

Lon: 62.800E

Elev: 185

StdP: 99.12

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-7.2	-15.5	1.0	-8.8	-12.6	1.3	-3.3	10.2	13.4	9.2	10.4	1.8	110	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	42.9	21.9	41.5	21.6	40.2	21.1	24.1	40.0	23.1	39.1	22.2	38.1	4.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.5	34.2	17.0	12.4	32.7	15.8	11.4	31.9	73.3	40.1	69.4	39.2	66.1	38.2	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.4	DB	-13.8	45.3	4.8	1.1	-17.2	46.1	-20.0	46.8	-22.7	47.4	-26.2	48.2	(4)
(5)			WB	-14.1	24.6	4.7	2.2	-17.5	26.2	-20.3	27.5	-23.0	28.8	-26.4	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	4.0	6.0	12.2	18.7	25.2	30.2	32.3	30.3	23.9	16.6	9.2	4.7	(6)	
	DBStd	11.05	5.51	5.74	5.18	5.00	3.71	3.04	2.54	3.09	3.82	4.97	5.39	4.96	(7)	
	HDD10.0	623	196	132	37	2	0	0	0	0	0	8	75	174	(8)	
	HDD18.3	1839	444	344	200	56	2	0	0	0	4	91	276	421	(9)	
	CDD10.0	3493	10	21	104	265	471	606	691	631	418	213	52	12	(10)	
	CDD18.3	1666	0	0	9	68	215	356	433	372	172	38	3	0	(11)	
	CDH23.3	25974	1	4	128	873	3019	5798	7264	5836	2408	587	54	1	(12)	
(13)	CDH26.7	16382	0	0	36	381	1692	3840	5003	3877	1314	223	16	0	(13)	
(14)	Wind	WSAvg	2.9	3.0	3.2	3.2	3.0	2.9	3.1	3.4	3.1	2.5	2.3	2.4	2.8	(14)
(15) Precipitation	PrecAvg	118	19	21	27	14	8	1	0	0	1	4	14	12	(15)	
	PrecMax	188	57	72	92	38	38	7	3	1	11	26	32	41	(16)	
	PrecMin	42	2	0	2	0	0	0	0	0	0	0	1	0	(17)	
	PrecStd	36	13	19	18	11	9	2	1	0	3	6	10	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	23.2	30.7	37.2	40.3	43.6	45.1	43.9	39.9	35.3	28.7	21.4	(19)	
		MCWB	10.8	12.6	16.0	19.6	21.2	22.0	22.2	22.9	20.7	18.5	15.5	10.9	(20)	
	2%	DB	17.4	19.9	26.8	33.6	38.2	41.8	43.4	42.1	37.7	31.5	24.3	18.3	(21)	
		MCWB	9.2	10.9	14.9	18.6	20.2	21.6	21.6	21.9	19.9	17.2	13.6	9.8	(22)	
	5%	DB	14.5	17.1	23.9	31.0	36.5	40.4	42.0	40.6	35.7	29.0	21.1	15.2	(23)	
		MCWB	8.2	9.9	13.6	17.3	19.5	21.0	21.6	21.4	19.2	16.3	12.0	8.6	(24)	
	10%	DB	11.8	14.7	21.1	28.4	34.6	39.0	40.4	39.0	33.6	26.6	18.4	12.6	(25)	
		MCWB	6.9	8.9	12.6	16.2	18.7	20.6	21.3	20.8	18.3	14.9	11.0	7.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	13.2	17.0	21.2	23.2	25.0	26.0	24.5	22.6	19.9	16.5	11.9	(27)	
		MCDB	19.0	20.3	27.8	33.7	37.8	40.7	42.0	40.8	37.1	33.0	25.8	19.4	(28)	
	2%	WB	10.1	11.9	15.6	19.5	21.7	23.4	24.3	23.5	21.2	18.0	14.4	10.4	(29)	
		MCDB	15.9	18.1	24.6	31.2	35.4	38.9	40.2	40.2	35.7	29.7	22.2	16.9	(30)	
	5%	WB	8.7	10.6	14.4	18.0	20.5	22.4	23.2	22.5	19.9	16.6	12.8	9.1	(31)	
		MCDB	13.5	16.0	22.5	28.4	33.4	38.2	39.5	39.0	34.1	27.6	19.8	14.2	(32)	
	10%	WB	7.2	9.2	13.1	16.9	19.4	21.4	22.2	21.5	18.6	15.3	11.4	7.8	(33)	
		MCDB	11.5	14.2	20.0	26.8	32.2	37.2	38.4	37.7	32.5	25.6	17.5	12.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	11.0	12.2	13.6	14.9	15.1	14.5	15.4	16.8	16.3	13.0	10.3	(35)	
		MCDBR	14.3	14.9	17.0	17.6	17.8	17.0	16.7	16.8	19.1	18.9	16.9	14.8	(36)	
		MCWBR	7.5	7.3	7.0	7.1	7.1	7.2	6.7	7.8	9.5	9.7	8.7	8.0	(37)	
	5% WB	MCDBR	12.7	13.1	15.1	15.5	15.6	15.2	15.1	15.7	17.5	17.4	15.1	13.6	(38)	
		MCWBR	7.1	6.9	6.6	6.6	7.3	7.8	7.9	8.4	9.9	9.4	8.0	7.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.389	0.433	0.506	0.535	0.534	0.542	0.521	0.476	0.450	0.474	0.440	0.392	(40)	
	taud		2.089	1.979	1.832	1.804	1.827	1.802	1.853	1.949	1.980	1.901	1.971	2.072	(41)	
	Ebn at Noon		752	767	748	756	766	759	771	797	790	711	680	711	(42)	
	Edh at Noon		119	151	193	210	209	215	202	179	164	161	133	113	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.21	3.17	4.33	5.72	7.05	7.97	7.85	7.25	5.99	4.20	2.71	2.03	(44)	
	RadStd		0.20	0.33	0.32	0.44	0.27	0.23	0.17	0.17	0.15	0.24	0.24	0.22	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.58	-1.16	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	-0.54	-0.48	N/A	N/A	N/A	+90	(47)

Nomenclature: See separate page